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Compact Lithium-Niobate-on-Insulator Polarization Rotator Based on Asymmetric Hybrid Plasmonics Waveguide

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Abstract: We proposed a compact and efficient polarization rotator (PR) based on an asymmetric hybrid plasmonics waveguide (HPW) on the lithium-niobate-on-insulator (LNOI) platform. We focus on the design of rotation angle and balance between the two hybrid modes. The presented device exhibits a polarization extinction ratio as high as 59.58 dB and an insertion loss of 1.44 dB with the rotation length of only 13.7 μm. The fabrication tolerances and a suitable process are also discussed to meet the state-of-art technology.

Index Terms: Lithium-niobate-on-insulator, polarization rotator, plasmonics.

1. Introduction

Lithium-niobate-on-insulator is an ideal platform for photonic integrated circuit (PIC) because of its outstanding electro-optical property, high second-order optical nonlinearity and acousto-optic property. Meanwhile, its high refractive index and small footprint compared with the bulk lithium niobate (LN) lead to enormous application potential with complementary metal-oxide semiconductor (CMOS) fabrication process [1]. However, due to the high index and inherent birefringence, LNOI always has trouble with the polarization sensitivity. In a polarization diversity system [2], [3], polarization rotators are of great importance to realize efficient polarization conversion.

Recently, various approaches are used to design on-chip PRs. Mode-evolution-based [4]–[7] PRs are not highly wavelength sensitive, which is a good choice to realize PR fabrication. but they require a quite long length, which makes it limits to meet the requirement of compactness. Besides, they usually had several layers and fine features, which increase the fabrication difficulties. Asymmetrical directional couplers [8]–[12] are also broadly adopted because most of them only need one-step etch. Nevertheless, they require different top cladding material to break the symmetry of the couplers, which lead to large footprints and complex fabrication processes. The mode-interference-based PRs show the potential to simultaneously solve the volume and fabrication problems of mode-evolution-based PR and asymmetrical directional coupling-based PR. The mode-interference-based PRs [13]–[18] can be much smaller, and the key point is the

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asymmetric waveguides to obtain rotated optical axes. There are several existing methods to build asymmetric waveguides, such as slanted waveguides [13], narrow trenches [14], asymmetric etched waveguides [16], and plasmonics waveguides [17], [18]. However, the first three methods have either long lengths or complex fabrication processes. In this case, the inherent advantages of natural polarization sensitivity and great ability to confine the light at nano-scaled structures makes the plasmonics attractive and mostly interested [19]. The plasmonics has been widely used in various polarization control devices, such as polarizers [20], [21] , polarization beam splitters [22], [23], mode converters [24] and mode-selective reflectors [25] . Recent reported mode-interference-based PRs on Silicon-on-Insulator (SOI) [17] and InP platform [18] showed that hybrid plasmonics waveguides have excellent properties of both light confinement and low propagation loss. Based on these, HPWs can be a suitable structure for mode-interference-based PRs. Unfortunately, the mode-interference-based PRs based on plasmonics waveguide on LNOI are seldom considered.

In this paper, we propose a mode-interference-based PR with an asymmetric HPW on LNOI, which features the advantages of compactness and efficiency. The optical axes of two hybrid modes (HMs) are rotated by the asymmetry metal layer, at the same time, forming different propagation constants of the two HMs. We calculate and adjust the total propagation loss difference of two HMs to optimize interference. The device shows a high polarization extinction ratio of 59.58 dB, which is the highest result to the best of our knowledge, and a low insertion loss of 1.44 dB with the interference length of 13.7 μm. Furthermore, we design a feasible fabrication process, and discuss the fabrication tolerance.

2. Device structure and design

The device contains input, output and rotation sections. Fig. 1 (a) and (b) show the three-dimensional view and the cross-section view of the proposed PR, respectively. The input and output sections are formed by the LN rib waveguides. The LN slab height, LN rib height, and LN rib width are hs, hr and wr, respectively. The total height of the LN $(hs + hr)$ is 600 nm, which is easy to obtain commercially. To meet the state of etching ability, the simulated sidewall angle is set to 70°, and hr is set to 300 nm. The rotation section is composed of a SiO_2 gap and an asymmetric Ag cap with the thickness of 100 nm. The thickness of the SiO_2 gap and Ag cap are hg and ha =100 nm, respectively. The left and right edges positions of Ag cap are determined by their Y coordinates ye and ya, respectively. The left edge of Ag cap is extended far enough (ye should be less than -1 μm) that the left edge has almost no effect on the waveguide. As shown in Fig. 1 (a), we choose the x-cut LNOI wafer, whose extraordinary refractive index (n_e) is 2.138 along the y-axis and ordinary refractive index (n_o) is 2.211 along the x- and z-axis at the wavelength of 1550 nm. The refractive indices of SiO_2 and Ag are 1.44 and 0.1453+11.3547i [26] at the wavelength of 1550 nm, respectively.

The finite difference eigenmode (FDE) solver is utilized to analyze the HMs in the HPW. The Ag cap breaks the structure's symmetry, leading to two HMs, HM_1 and HM_2, as shown in Fig. 2. Compared with the SOI platform [17], LNOI has a lower refractive index, resulting in a larger mode area and larger device length. In order to lessen the PR, we consider utilizing the inclined sidewall and excite the hybrid plasmonics mode at the sidewall. It makes a stronger optical axis rotation than that of only putting the Ag cap on the top of the rib. We put one edge of the Ag cap on the waveguide, and extend the other edge far enough (the Y coordinates less than -1 μm) from the waveguide to avoid extra plasmonics modes. The two HMs have optical axes with 45° rotation to the z-axis. The length of the rotation waveguide L is calculated as:

$$L = L_c = \frac{\pi}{\beta_1 - \beta_2} = \frac{\pi}{(n_{eff1} - n_{eff2})\, k_0} \tag{1}$$

where L_c is the half interference length, β_1 and β_2 are the propagation constants of the two HMs, n_{eff1} and n_{eff2} are the effective refractive indices of the two HMs, and k_0 is the vacuum wave

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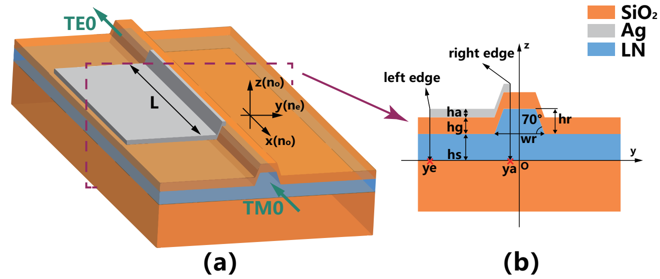

Fig. 1. Schematic of the PR. (a) three-dimensional view and (b) cross-section view.

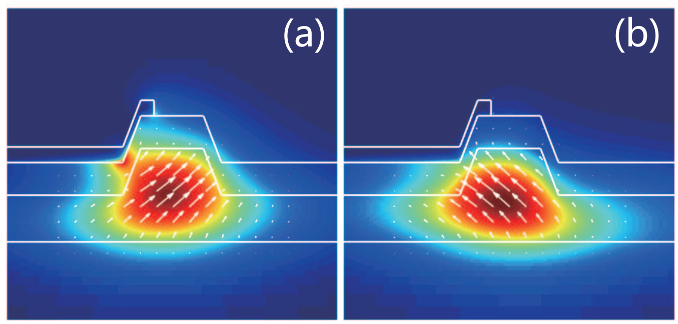

Fig. 2. Magnetic field profile of two hybrid modes. (a) HM_1 and (b) HM_2. The arrow is the direction of magnetic field. Here, $hg = 200$ nm, $ha = 100$ nm and $ya = -110$ nm.

number. Assuming a fundamental transverse magnetic mode (TM_0) injected, it will excite the two HMs in the rotation waveguide. The two HMs have a difference in propagation constants, leading to a phase difference during propagating. Then the two modes beat with each other with a π-phase difference after traveling through the rotation section. Then, the two HMs couple into one mode, which is rotated by 90° compared with the input mode. In other words, the input TM_0 mode has become the fundamental transverse electric mode (TE_0).

The key figure of merit for the PRs is the polarization conversion efficiency (PCE), which is defined as the ratio of the desired polarization to the sum of both polarizations at the output port. For interference-based PRs, PCE is related to the optical axis rotation angle of the two HMs [13] :

$$\text{PCE} = \sin^2(2\theta)\sin^2\left(\frac{\pi L}{2L_c}\right) \tag{2}$$

Here, the rotation angle θ is defined as [13] :

$$\tan(\theta) = \frac{\iint \varepsilon(y,z)\, E_y^2(y,z)\, dydz}{\iint \varepsilon(y,z)\, E_z^2(y,z)\, dydz} \tag{3}$$

where $\varepsilon(y,z)$ is the real part of the permittivity distribution, $E_y(y,z)$ and $E_z(y,z)$ are the transverse and horizontal electrical components of the HMs, respectively. It should be noted that the $\varepsilon(y,z)$ should be replaced by $\varepsilon_y(y,z) = n_y^2$ and $\varepsilon_z(y,z) = n_z^2$ for the sake of the anisotropy of LN. Obviously, $\theta = 45°$ is needed to obtain a 100% PCE.

As for an interference structure, it is important to keep the balance of two HMs, which directly affects the polarization extinction ratio (PER) and insertion loss (IL). The balance of two HMs can be broken by the rotation angle θ and the different mode loss of the two HMs.

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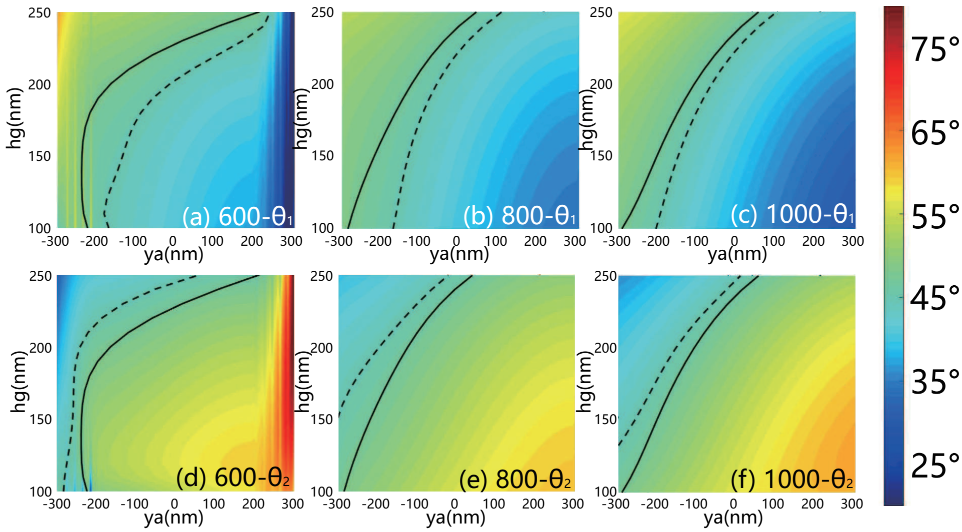

Fig. 3. Calculated rotation angle θ with different hg and ya. (a) $wr = 600$ nm, θ_1 (b) $wr = 800$nm, θ_1 (c) $wr = 1000$ nm, θ_1 (d) $wr = 600$ nm, θ_2 (e) w$r = 800$nm, θ_2 (f) w$r = 1000$nm, θ_2 The black solid lines show the position where $\theta_1 = \theta_2$. The black dotted lines show the position where (a)-(c) $\theta_1 = 45°$ and (d)-(f) $\theta_2 = 45°$.

The rotation angle θ affects both the PCE, and the balance of two HMs. For the specified hr and hs, the TM_0 mode was cut off at $wr \leq 570nm$. In this paper, we consider wr as 600 nm, 800 nm, and 1000 nm to research the HMs. The rotation angle is related to wr, hg and ya. We calculate θ at a region of hg and ya with different wr, as shown in Fig. 3, where θ_1 and θ_2 are the rotation angle of HM_1 and HM_2, respectively. Intuitively, as hg and wr increase, it needs a larger ya to obtain $\theta = 45°$. For the balance of the two HMs, $\theta_1 = \theta_2$ is required to equalize the rotation abilities of the two HMs. Unfortunately, for the specified hr and hs, the points of $\theta_1 = \theta_2$ cannot achieve $\theta = 45°$. Next, we calculated the PCE by Eq. (2), the result shows that when $|\theta - 45°| \leq 2.6°$, the maximum PCE can be over 99.1%, which is acceptable to achieve a high PER at the cost of a small IL.

The mode losses include two main parts: propagation losses and coupling losses. The propagation losses are the intrinsic losses of HMs, and the coupling losses include input coupling losses and output coupling losses. The intrinsic losses are caused by metal absorption. The mode intrinsic losses and the loss difference between HM1 and HM2 are calculated in the case of $\theta = 45°$, as shown in Fig. 4. It is noted that the mode intrinsic loss of HM_1 is always larger than that of HM_2. The difference of losses will cause different output powers of the two HMs, which weakens the interference. Fig. 4 (b) shows the interference length L_c and the loss difference $\Delta loss$ between HM_1 and HM_2. $\Delta loss$ is calculated with $\Delta loss = (loss_1 - loss_2)\, L_c$, where $loss_1$ and $loss_2$ are the mode intrinsic loss of the HM1 and HM2, respectively. L_c increases and $\Delta loss$ decreases with wr and hg increase. When hg >200 nm, wr has a little impact on $\Delta loss$. This is because, when hg is larger, the metal is more far away from the LN rib waveguide, thus its influence becomes weaker. As a result, the n_{eff} (effective refractive indices) of the two HMs become similar and the metal absorption decrease, reducing the smaller $\Delta loss$. Therefore, wr and hg should be as small as possible in the case of wr ≥570 nm and hg >200 nm, to keep both L_c and $\Delta loss$ small.

The coupling loss is calculated by the overlap among TE_0, TM_0 and HMs, as shown in Fig. 5. The overlap is defined as:

$$overlap = \left| \mathrm{Re}\left[\frac{\left(\int \mathbf{E}_1 \times \mathbf{H}_2^* \cdot d\mathbf{S}\right)\left(\int \mathbf{E}_2 \times \mathbf{H}_1^* \cdot d\mathbf{S}\right)}{\left(\int \mathbf{E}_1 \times \mathbf{H}_1^* \cdot d\mathbf{S}\right)} \right] \frac{1}{\mathrm{Re}\left(\mathbf{E}_2 \times \mathbf{H}_2^* \cdot d\mathbf{S}\right)} \right| \quad (4)$$

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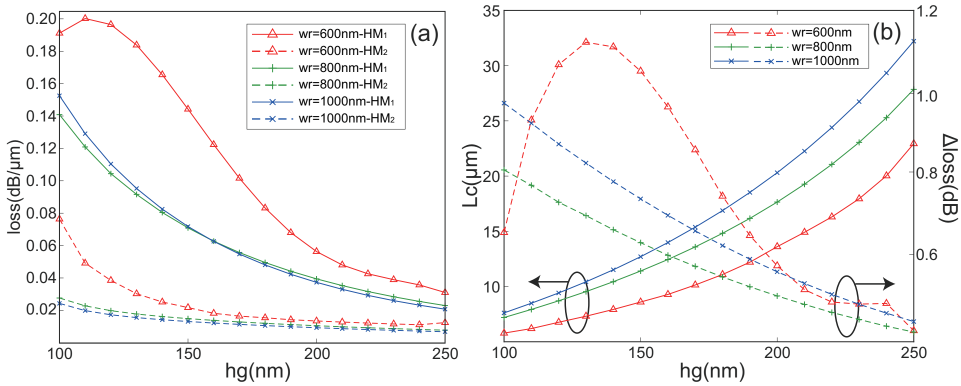

Fig. 4. (a) Mode intrinsic losses of the two HMs and (b) interference length L_c (solid lines) and loss difference ($\Delta loss$) between HM$_1$ and HM$_2$ (dotted lines).

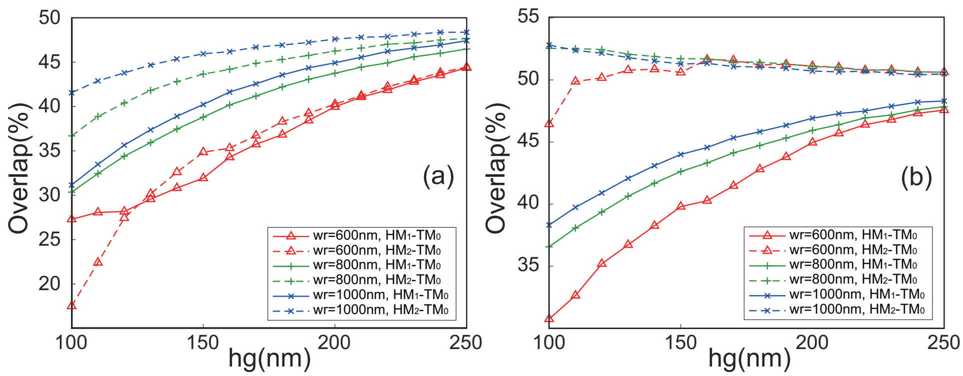

Fig. 5. The overlap among HMs and (a) TM$_0$ and (b) TE$_0$.

The ideal overlap of the HMs is 50% to obtain a lossless coupling. However, the irregular shapes of the HMs lead to coupling loss. The input coupling (HMs to TM$_0$) difference mainly affects PER, and both input and output coupling (HMs to TE$_0$) affect IL. As shown in Fig. 5 (a), though the overlap between HMs and TM$_0$ increases with the increase of wr, a small wr shows a small input coupling difference. When wr = 600 nm, the IL is about 0.5 dB higher than that of wr = 800 nm and wr = 1000 nm, but the overlap of two HMs is almost the same, which means a very high PER. Moreover, the output coupling of various wr are almost the same when 200 nm. As shown in Fig. 5 (b), when hg >200 nm, wr has a little impact on the overlap between HMs and TE$_0$.

To obtain the best PER, we calculate the total propagation loss difference (TPLD) of the two HMs as:

$$TPLD = \left| overlap_{\text{HM1 - TM0}} \cdot 10^{\left(-\frac{Lc\cdot loss_1}{10}\right)} - overlap_{\text{HM2 - TM0}} \cdot 10^{\left(-\frac{Lc\cdot loss_2}{10}\right)} \right| \tag{5}$$

The calculated results are shown in Fig. 6. It can be seen that, TPLD tends to be 0 when wr =600 nm and hg >200 nm because both $\Delta loss$ and overlap are at a very low level. For the sake of high PER, low IL and compactness, wr =600 nm and hg =210 nm are chosen. The rest of the parameters are calculated as ya =-110 nm and L_c = 14.8 μm.

3. Results and discussion

In order to verify the design principles and demonstrate the expected performance, the three-dimensional finite difference time domain (3D-FDTD) method is utilized to calculate the light propagation in the proposed PR numerically. In the simulation, a TM$_0$ (z-polarized) source is

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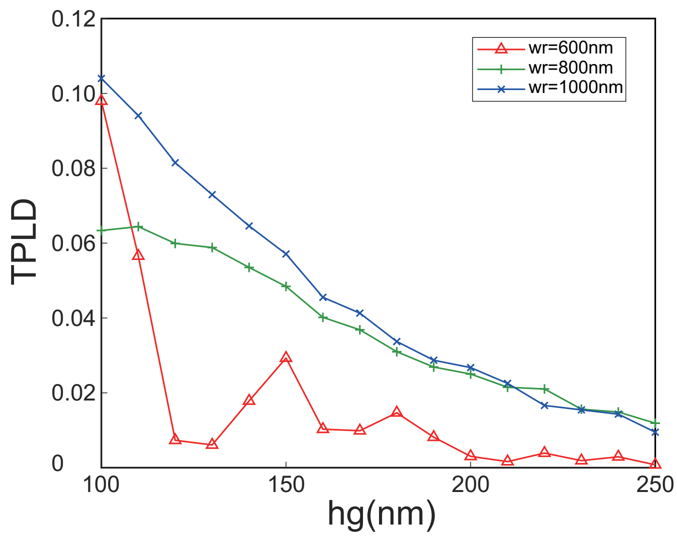

Fig. 6. TPLD as a function of hg.

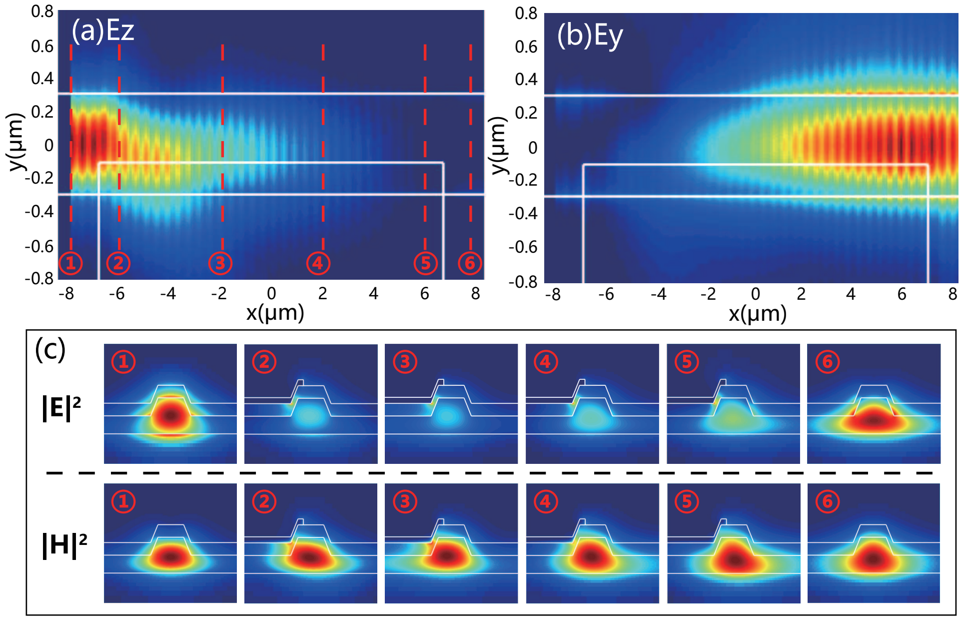

Fig. 7. The electric field propagation profiles of the proposed PR at 1.55μm wavelength for (a) E_z , (b) E_y and (c) the electric-field and magnetic-field distributions in different locations when TM$_0$ mode input from the left port.

launched at the left port. The corresponding electric field propagation profiles and transmission spectra are shown in Fig. 7. One sees that when TM0 mode is launched, E_z component fades away in the rotation section, and E_y begins to appear and becomes dominant in the output section.

The transmission spectra are calculated as shown in Fig. 8, when input power is normalized in 1 (0dB), the PER and IL are defined as [17]:

$$\begin{aligned} PER &= 10\lg\left(P_{TE}/P_{TM}\right), \\ IL &= -10\lg\left(P_{TM}\right) \end{aligned} \tag{6}$$

The calculated PER and IL at the wavelength of 1548.9 nm are 59.58 dB and 1.44 dB, re-

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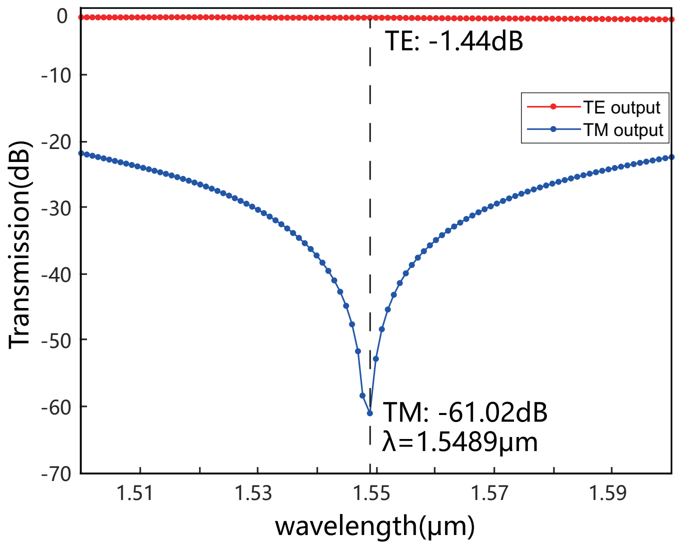

Fig. 8. Calculated transmission spectra at the output port for TE_0 and TM_0. Here, $wr = 600$ nm, $ya = -120$ nm, $hg = 210$ nm and $L = 13.7\mu$m.

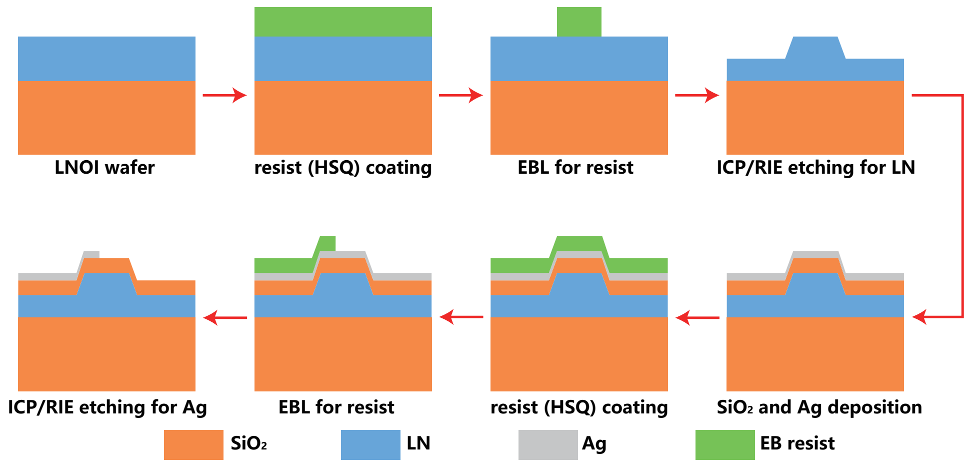

Fig. 9. Design of fabrication process of the proposed PR.

spectively. The bandwidth is 31.3 nm for PER>30 dB, which is above 100 nm for PER>20 dB. Besides, we find that the best interference length is $L = 13.7\mu$m instead of the calculated 14.8 μm, and the best ya is -120 nm instead of the calculated -110 nm. The partial reason is because the small differences between the two simulation methods. The boundaries between the rotation section and input and output waveguides lead to an extra plasmonics mode along the Ag edges, which is ignored by the FDE solver, resulting in a deviation of the equivalent rotation length and rotation angle.

In addition, we design a feasible fabrication process as shown in Fig. 9. The proposed PR is fabricated by electron beam lithography (EBL), deposition and ICP/RIE etching. Hydrogen silsesquioxane (HSQ) is utilized as the EB resist. It should be emphasized that although Ag is not compatible with CMOS processing, electron beam (EB) evaporation can be used to form Ag films on SiO_2 [27], [28]. The fabrication process only needs two-step etching, resulting in a small accumulated error.

The fabrication tolerance is also analyzed. Fig. 10 shows the calculated transmission spectra as

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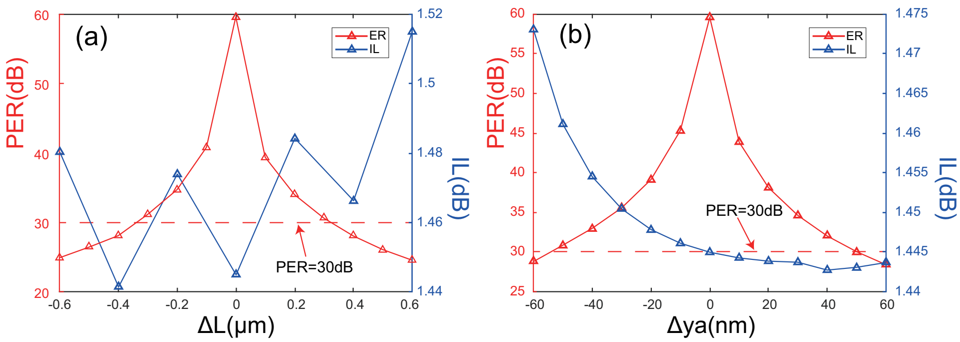

Fig. 10. Fabrication tolerance of the PR with the deviation of (a) L and (b) ya.

a function of rotation length variation (ΔL) and Ag edge position deviation (Δya) at the wavelength of 1548.9 nm. It can be seen that the PR exhibits a large tolerance to the fabrication error. The PER is greater than 30 dB at the range of -0.3μm $< \Delta L < 0.3\mu$m and -50nm $< \Delta ya < 50$nm, with IL varying in the range of 0.08 dB and 0.04 dB, respectively. The simple fabrication process and large fabrication tolerance ensure our PR easy to fabricate.

4. Conclusion

In summary, a mode-interference-based polarization rotator is demonstrated with an asymmetric HPW on the LNOI platform. The rotation angle and propagation loss difference are properly designed, resulting in the polarization extinction ratio as high as 59.58 dB. The insertion loss is only 1.44 dB with a rotation length of 13.7 μm. The proposed PR shows a broadband performance maintaining ER$>$30 dB in the bandwidth range of 30 nm. The structure is easy to design and fabricate due to a simple structure and fabrication process. Moreover, the proposed PR is designed based on the conventional LNOI waveguide, leading to huge potential as a building block in a compact on-chip LNOI polarization diversity scheme.

Acknowledgements

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